

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

1. Functional Requirements

S.No	Requirement ID	Requirement Description	Priority	Status
1	FR-01	System shall accept	Must Have	Done
2	FR-02	System shall classify the input into one of 5 emotions: Bored,		

Confident, Confus

			Must Have	Done
3	FR-03	System shall display the	Must Have	Done
4	FR-04	System shall display		
		per-emotion probability		

scores as a bar chart

			Must Have	Done
5	FR-05	System shall detect		

mixed emotions when

probability gap between

top 2 emotions <

Should Have	Done
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S.No	Requirement ID	Requirement Description	Priority	Status
6	FR-06	System shall apply	Should Have	Done
7	FR-07	System shall generate	Must Have	Done
8	FR-08	System shall log every	Should Have	Done
9	FR-09	System shall display an	Could Have	Done
10	FR-10	System shall display a	Could Have	Done

model comparison page

2. Non-Functional Requirements

S.No	Requirement ID	Description	Target
1	NFR-01	**Performance** - Emotion	< 5 seconds
2	NFR-02	**Accuracy** - Validation set	80%
3	NFR-03	**Availability** - App uptime on	99% (free tier)
4	NFR-04	**Usability** - No login or	

technical knowledge required

Zero friction

S.No	Requirement ID	Description	Target
5	NFR-05	**Scalability** - App must	10 simultaneous users
6	NFR-06	**Security** - Gemini API key	Stored in Streamlit Secrets
7	NFR-07	**Compatibility** - Works on	All major browsers
8	NFR-08	**Maintainability** - Modular	PEP-8 compliant
		codebase with clear folder	

3. System Constraints

Constraint	Description
Deployment	Streamlit Community Cloud (free tier) - no TensorFlow/PyTorch on cloud
API Limit	Google Gemini free tier: 1500 requests/day
Model Serving	Cloud mode uses Gemini for detection (no local model weights)
Data Privacy	No user data stored beyond session-scoped CSV on the server
Language	English only (input language constraint)

4. User Stories

ID	As a...	I want to...	So that...
US-01	Student	Type my study problem	I can get an accurate emotion detection
US-02	Student	See my detected emotion with confidence	I understand how the system interpreted my text

ID	As a...	I want to...	So that...
US-03	Student	Get personalized AI guidance	I can overcome my learning struggle
US-04	Student	View session history	
US-05	Educator	View analytics dashboard	I understand overall student emotional trends
US-06	Developer	Train custom models	